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PERFORMANCE SPECIFICATION

NOZZLE, PROBE, AERIAL REFUELING, TYPE MA-2

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1. SCOPE

1.1 Scope. This specification covers aerial refueling probe nozzles and lock rings intended for use on receiver aircraft equipped with probe type aerial refueling systems.

1.2 Classification. The probe nozzles covered under this specification are type MA-2 and incorporate a flexible tip.

1.2.1 Types.

a. Type 1: (Obsolete) Nozzle capable of off-center disconnects up to 15 degrees maximum.

b. Type 2: Nozzle capable of off-center disconnects up to 22.5 degrees maximum.

1.2.2 Classes.

a. Class A: (Obsolete) Nozzle capable of operation in the -67 to 160 °F ambient temperature range and fuel temperature range of -67 to 135 °F.

b. Class B: Nozzle capable of operating in the -67 to 350 °F ambient temperature range and fuel temperature range of -67 to 200 °F.

2. APPLICALBE DOCUMENTS

2.1 General. The documents listed in this section are referenced in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification, recommended for additional information, or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-25988	-	Rubber, Fluorosilicone Elastomer, Oil-, Hydraulic Fluid-, and Fuel-Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes

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MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-I-8500	-	Interchangeability and Replaceability of Component Parts for Aerospace Vehicles
MIL-PRF-7024	-	Calibrating Fluids, Aircraft Fuel System Components
MIL-PRF-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-PRF-81975	-	Couplings, Regulated, Aerial Pressure Refueling Type MA-2, Type MA-3 and Type MA-4
MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series: General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-464	-	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Dissimilar Metals
MIL-STD-31000	-	Technical Data Packages

(Copies of these documents are available online at https://assist.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent

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specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

PUBLICATIONS

NORTH ATLANTIC TREATY OGRANIZATION (NATO)

ATP 3.3.4.6 (STANAG 3447)	-	Air-to-Air (Aerial) Refueling Equipment: Probe-Drogue Interface Characteristics

(Copies of these documents are available online at https://coi.japcc.org/aar/)

MILITARY STANDARDS

MS24354	-	Drogue Cone, Nozzle and Reception Coupling – Type MA-2 Flight-Pressure-Refueling System, Assembly of
MS24356	-	Detailed Specification Sheet, Nozzle, Probe, Aerial Refueling, Type MA-2
MS24359	-	Nose, Probe Mast – Type MA-2, Flight Pressure Refueling Nozzle, Outline Dimensions for
MS24360	-	Detailed Specification Sheet, Nozzle, Probe, Aerial Refueling, Type MA-2

(Copies of these documents are available online at https://assist.dla.mil.)

2.3 Non-Government documents. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM20995	-	Wire, Safety or Lock
NASM25027	-	Nut, Self-Locking, 250 °F, 450 °F, and 800 °F
NASM33540	-	Safety Wiring, Safety Cabling, Cotter Pinning, General Practices for

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NASM33588	-	Nut, Self-Locking, Aircraft, Reliability and Maintainability Usage Requirements for

(Copies of these documents are available online at https://www.aia-aerospace.org/.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1	-	Surface Texture (Surface Roughness, Waviness, and Lay)
ASME Y14.100	-	Engineering Drawing Practices

(Copies of these documents are available online at https://www.asme.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1655	-	Standard Specification for Aviation Turbine Fuels
ASTM D5900	-	Standard Specification for Physical and Chemical Properties of Industry Reference Materials (IRM)

(Copies of these documents are available online at https://www.astm.org.)

GOSUDARSTVENNYY STANDART, means STATE STANDARD (GOST-Russian)

GOST 10227 (TS-1)	-	Jet Fuels Specifications

(Copies of this document are available from GOST Interstandard, 117049 Moscow, B-29 Russia, Attn: Mr. Nickolay Usov, nusov@gost.ru or in English from https://ewb.ihs.com/.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS568	-	Aerospace Size Standard for O-rings
SAE AS8879	-	Screw Threads - UNJ Profile, Inch, Controlled Radius Root with Increased Minor Diameter
SAE AS83485	-	O-Ring Molded from AMS7287 Material
SAE AMS-QQ-416	-	Plating, Cadmium (electrodeposited)

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SAE AMS1428	-	Fluid, Aircraft Deicing/Anti-Icing, Non-Newtonian (Pseudoplastic), SAE Types II, III, and IV
SAE AMS2175	-	Castings, Classification, and Inspection of
SAE AMS2447	-	Coating, Thermal Spray High Velocity Oxygen/Fuel Process
SAE AMS3384	-	Rubber, Fluorocarbon Elastomer (FKM) 70 to 80 Hardness, Low Temperature Sealing Tg -22 °F (-30 °C) For Elastomeric Shapes or Parts in Gas Turbine Engine Oil, Fuel and Hydraulic Systems
SAE AMS7259	-	Rubber: Fluorocarbon (FKM) High Temperature/Fluid Resistant Low Compression Set/ 85 to 95 Hardness For Seals In Fuel Systems and Specific Engine Oil Systems
SAE AMS7287	-	Fluorocarbon Elastomer (FKM) High Temperature / HTS Oil Resistant / Fuel Resistant Low Compression Set / 70 to 80 Hardness, Low Temperature Tg -22 °F (-30 °C) For Seals in Oil / Fuel / Specific Hydraulic Systems
SAE ARP868	-	Method-Pressure Drop Tests for Fuel System Components

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Assembly. The nozzle consists of an assembled unit conforming to the dimensions shown on Figure 1.

Note: Attachment parts (O-ring per MIL-DTL-25988/1B, and MS24360 (shown on Figure 2) are not considered a portion of the nozzle. Installation features of these attachments are shown in MS24354. The basic components of a nozzle and their typical locations are shown on Figure 3.

3.2 Materials.

3.2.1 Fluid resistance. All materials used in the construction of the nozzle shall be fuel resistant when tested in any sequence with fluids conforming to the following specifications:

a. ASTM D910
b. MIL-DTL-5624 (JP4 & JP5)
c. MIL-PRF-680
d. ASTM D5900, IRM 901 and IRM 903
e. MIL-PRF-7024, Type II

3.2.2 Corrosion-resistant material. All materials shall be corrosion resistant or treated to resist corrosion from fuels, salt spray, and atmospheric conditions to be encountered in storage or normal service.

3.2.2.1 Protective coatings. Protective coatings shall not chip, crack, or flake. Cadmium plating, when used, shall conform to SAE-AMS-QQ-P-416. No painted surfaces are allowed.

3.2.2.2 Aluminum protection. Aluminum alloy parts shall be anodized in accordance with MIL-PRF-8625, or EN4827, or treated as approved by the qualifying activity.

3.2.3 Prohibited materials. Magnesium, chromium, zinc, and copper or their alloys shall not be used.

3.2.3.1 Impact materials. PTFE (polytetrafluoroethylene) shall not be used in any location where impact loads may be encountered.

3.2.3.2 Age-sensitive materials. Age sensitive materials shall not be used, except those specified in 3.4.1.

3.2.4 Castings. Castings shall meet or exceed Class 3, Grade C requirements as specified in SAE-AMS2175.

3.2.5 Dissimilar metals. Dissimilar metals shall not be used in contact with each other, as specified in MIL-STD-889, unless protected against electrolytic corrosion.

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3.2.6 Electrical protection.

3.2.6.1 Electrical bonding. The nozzle interface to the probe mast (MS24359) shall be capable of achieving an electrical bonding resistance value of 1.0 ohm or less (Class S bond, per MIL-STD-464) between the nozzle body and the probe mast after installation per MS24354.

3.2.7 Synthetic rubber parts. Equipment or assemblies containing synthetic rubber parts shall be marked in accordance with MIL-STD-190 or equivalent.

3.3 Selection of specification and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected by standard commercial practices.

3.4 Construction. The nozzle shall be designed to prevent parts from working loose during shipment, storage, or operation use; and withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and use.

3.4.1 O-rings, packings, and seals. O ring, packing, and seal design and construction shall ensure maximum protection against leakage and damage due to friction and conform to SAE AS568, MIL-DTL-25988/1B, SAE AS83485, SAE AMS3384, SAE AMS7287, SAE AMS7259, or as approved by the qualifying activity.

3.4.2 Threaded parts. Screw threads shall conform to MIL-S-7742 or SAE AS8879 or as approved by the qualifying activity.

3.4.2.1 Prohibited threads. Pipe threads shall not be used.

3.4.2.2 Locking of threaded parts. All threaded parts shall be positively locked by safety wiring, self-locking nuts, or other methods as approved by the qualifying activity.

3.4.2.2.1 Safety wiring. Threaded parts utilizing safety wiring shall conform to NASM20995 and be installed in accordance with NASM33540.

3.4.2.2.2 Self-locking nuts. Threaded parts utilizing self-locking nuts shall conform to NASM25027 and be used in accordance with NASM33588.

3.4.2.2.3 Prohibited locking. The use of lock washers, set screws, cotter pins, or staking shall not be permitted unless specifically approved by the qualifying activity.

3.4.3 Weight. The maximum dry weight of the nozzle shall not be greater than 5 pounds.

3.4.4 Finishes. The nozzle finish shall be smooth and free from sharp edges.

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3.4.4.1 Surface roughness. Surface roughness, where indicated on Figure 1, shall be as specified in accordance with ASME B46.1.

3.4.5 Reliability. The nozzle shall be designed and constructed in a manner that will ensure the highest degree of operational reliability under all service conditions when operated with a test coupling.

3.4.6 Maintainability. The nozzle shall not require scheduled maintenance.

3.4.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable in accordance with MIL-I-8500.

3.4.8 Dimensions. The nozzle dimensions shall be in accordance with Figure 1.

3.4.9 Lubrication. The nozzle shall operate satisfactorily without requiring lubrication other than the fluid being transferred.

3.4.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130 or MIL-STD-190 as applicable.

3.4.10.1 Application. The identification shall be permanent and not peel off, crack, or loosen with age, environmental conditions, or damage resulting from aerial refueling operations.

3.4.10.2 Location. The location of markings is shown on Figure 3 and shall not affect the performance of the nozzle.

3.4.10.3 Markings. All nozzle types shall include, at a minimum: part number, serial number, and U.S. Government National Stock Number (NSN).

3.4.10.4 Part number. The item identification and part number requirements of ASME Y14.100 shall govern the manufacturer’s part numbers and changes thereto.

3.4.11 Workmanship. The nozzle shall be uniform in quality, clean, free from defects, parts properly aligned, tightly fitted together, and all burrs removed.

3.4.12 Incorrect assembly. The nozzle shall be designed to prevent incorrect assembly.

3.5 Operation.

3.5.1 Fuel transfer. The nozzle shall allow the passage of fuel when engaged with a test aerial refueling reception coupling.

3.5.2 Installation. The nozzle shall be capable of installation on probe masts designed in accordance with Figure 7 using lock ring (MS24360-2), Figure 2. Note that the

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aluminum lock ring (MS24360-1) is obsolete. Only the CRES lock ring (MS24360-2) is airworthy.

3.5.2.1 Nozzle sleeve operation. The sleeve latches shall prevent opening of the sleeve to tip seal until they are depressed flush with the nozzle body surface.

3.5.3 Sleeve latching mechanism. The nozzle shall have two sleeve latches spaced 180 degrees apart.

3.5.3.1 Engagement. The axial force required to depress the sleeve latches shall not be greater than 20 pounds in the engaging direction. The sleeve latches shall operate without binding when engaging the coupling.

3.5.3.2 Disengagement. The axial force required to depress the sleeve latches shall not be greater than 50 pounds in the disengaging direction. The sleeve latches shall operate without binding when disengaging the coupling.

3.5.4 External power source. The nozzle shall engage and disengage from the coupling without the use of any external source of power other than the forces generated by the receiver aircraft.

3.5.5 Off-center disconnects. When engaged with a test coupling, the nozzle shall disconnect without binding, continuous leakage, hanging up within the coupling, or breakage of any nozzle parts by a coupling disengaging force acting at angles up to and including 22.5 degrees from the longitudinal axis of the nozzle.

3.5.6 Emergency disconnect. The engaged nozzle shall withstand disconnects from a coupling pressurized with fuel up to 50 ±2 psig, at angles up to 22.5 degrees without binding or failure.

3.5.7 Proof pressure.

3.5.7.1 Engaged proof pressure. The nozzle, while engaged with a test coupling, shall withstand a minimum internal hydrostatic pressure of 240 psig with no external leakage, permanent deformation, structural failure, or binding of the nozzle sleeve.

3.5.7.2 Disengaged proof pressure. The nozzle, while not engaged with a test coupling, shall withstand a minimum internal hydrostatic pressure of 240 psig with no external leakage, deformation, or structural failure.

3.5.8 Ultimate (burst) pressure.

3.5.8.1 Engaged ultimate pressure. The nozzle while engaged with a coupling shall withstand a minimum internal hydrostatic pressure of 360 psig with no external leakage, or rupture. Following the test, the unit may be deformed and not function properly.

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3.5.8.2 Disengaged ultimate pressure. The nozzle while not engaged with a coupling shall withstand a minimum internal hydrostatic pressure of 360 psig with no external leakage, or rupture. Following the test, the unit may be deformed and not function properly.

3.5.9 Leakage and spillage.

3.5.9.1 Initial allowances. The initial allowable leakage and spillage shall be as follows:

a. When the engaged nozzle and coupling are subject to constant inlet pressures of 2 and 60 psig, there shall be no external leakage.

b. When the engaged nozzle and coupling are subject to an inlet fluid pressure of 50 ±2 psig and fluid flow of 100 to 600 gal/min, the spillage shall not be greater than 30 cc per disengagement.

c. When the nozzle is engaged and disengaged in a coupling, with an inlet fluid pressure of 10 psig, spillage shall not be greater than 25 cc per engagement or disengagement.

3.5.9.2 Subsequent allowances. The allowable leakage and spillage subsequent to significant coupling operations (as defined by the completion of the test sequence prior to the first “Leakage and Spillage (Subsequent)” test call-out in Table IV and Table V) shall be as follows:

a. When the engaged nozzle and coupling are subject to constant inlet pressures of 2 and 60 psig, there shall be no external leakage in excess of one drop per minute.

b. When the engaged nozzle and coupling are subject to an inlet fluid pressure of 50 ±2 psig and fluid flow of 100 to 600 gal/min, the spillage shall not be greater than 30 cc per disengagement.

c. When the nozzle is engaged and disengaged in a coupling with an inlet fluid pressure of 10 psig, spillage shall not be greater than 25 cc per engagement or disengagement.

3.5.9.3 Immersion leakage. The nozzle shall not allow fluid to enter the nozzle when the sleeve is in the closed position with an internal nozzle pressure of 2 psi (4 inHg) below ambient.

3.5.10 Pressure drop. The pressure drop through the nozzle engaged with a test coupling, using Grade JP-5 (or equivalent), shall not be greater than 8 psid at 600 gpm.

3.5.11 Contaminated fuel. The nozzle shall meet its functional requirements after exposure to contaminants specified in Table II.

3.5.12 Design limit loads.

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3.5.12.1 Limit loads. The nozzles shall be subjected to the following minimum loads without breakage, permanent deformation, or any other failure which would render it inoperable:

a. A 1,000-pound tensile load in combination with a 3,000-pound radial load.

b. A 1,000-pound compression load with a 3,000-pound radial load.

c. A 2,000-pound tensile load.

d. A 2,000-pound compression load.

3.5.12.2 Nozzle tip deflection. The nozzle tip shall be mounted in such a manner that when the sleeve is held in the fully open position, it is possible to deflect the tip in any radial direction until it contacts the sleeve lip without permanent deformation or structural failure. The nozzle tip shall return to its original, un-deflected position when released.

3.5.13 Nozzle-coupling compatibility. The nozzle shall be compatible with a test coupling such that it engages and disengages the coupling with no hang-ups (including rotation about the longitudinal axis) and meets the requirements of 3.5.3.1 and 3.5.3.2.

3.5.14 Environmental requirements.

3.5.14.1 Salt fog. The nozzle shall meet its performance requirements after exposure to salt fog in accordance with 4.3.5.14.1.

3.5.14.2 Dust. The nozzle shall meet its performance requirements after exposure to dust in accordance with 4.3.5.14.2.

3.5.14.3 Vibration. The nozzle shall meet its performance requirements after being exposed to the vibration requirements illustrated on Figure 5.

3.5.14.4 Impact and endurance.

3.5.14.4.1 Impact. When the nozzle is dropped as specified in Table III for impact, leakage shall not be greater than 25 cc per in-line engagement or disengagement.

3.5.14.4.2 Endurance. When the nozzle is dropped as specified in Table III for endurance, leakage shall not be greater than 25 cc per in-line engagement or disengagement and 100 cc per angular disengagement.

3.5.15 Extreme fuel and ambient temperature. The nozzle shall meet its functional requirements after exposure to extreme temperatures as specified in Table I.

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3.5.16 Disassembly and inspection. Parts shall not show any distortion, weakness, or abnormal wear after exposure to testing outlined in section 4.1.1.

3.5.17 Functional.

3.5.17.1 Disengaged leakage. When the nozzle alone is subjected to constant inlet pressures of 2 psig for one minute and 60 psig for one minute, there shall be no external leakage (initial) and no seepage in excess of one drop per minute (subsequent).

3.5.17.2 Engaged leakage. When the engaged coupling and nozzle are subjected to constant inlet pressures of 2 psig and 60 psig, there shall be no external leakage (initial) and no seepage in excess of one drop per minute (subsequent).

3.5.17.3 Opening sleeve force. The force required to depress the nozzle sleeve to the fully open position shall be 45 ±5 pounds.

3.5.17.4 Closing sleeve force. The nozzle sleeve shall achieve full closure with an opposing force of 15 pounds or greater.

3.5.17.5 Engagement and disengagement. The nozzle shall be capable of engagement and disengagement with a test coupling without evidence of binding, chattering, or excessive force.

4. VERIFICATION

4.1 Inspection/Tests. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities with the necessary equipment for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Qualification and inspection/tests. Qualification tests shall utilize two nozzles and consist of the complete series of tests and inspections specified below, performed in the order listed. If a different test order is desired, it shall be approved by the qualifying activity prior to test start. At the completion of testing, the qualification units shall be provided to the qualifying activity.

a. Inspection/Test Sequence No. 1 – A single qualification nozzle shall be utilized for the sequence of inspections/tests in Table IV.

b. Inspection/Test Sequence No. 2 – A single qualification nozzle shall be utilized for the sequence of inspections/tests in Table V.

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4.1.1.1 Qualification failures. If the part fails during qualification testing, the test shall be terminated, and the qualifying activity notified. The qualifying activity shall determine the retest requirements.

4.1.1.2 Samples for government testing. Additional test nozzles for Government or procuring agency testing shall be submitted to the qualifying activities as specified in the contract.

4.1.1.3 Government tests. At the option of the qualifying activity, the test nozzles that were submitted to the Government (see 4.1.1.2) may be subjected to flight, service, ground, and laboratory tests in accordance with the requirements of the qualifying activity.

4.1.1.4 Drawings. The contractor shall submit drawings conforming to ASME Y14.100, to the qualifying activity, with the test samples. The drawings shall include a cutaway section showing all parts in their normal assembled positions and shall specify the part numbers of all parts and subassemblies. The following data shall be furnished on or together with the assembly drawings:

a. Overall dimensions.

b. Materials, construction, treatments, and finishes.

c. Performance requirements and limitations.

d. Contractors and sub-vendors.

4.1.1.5 Test report. The test report shall be submitted to the qualifying activity with the two test samples. The report shall contain, as a minimum, the following:

a. Detailed reports on all tests, with copies of test log sheets, including the degree of conformance to the requirements of this specification.

b. Diagrams and photographs of all test setups and a listing of equipment used.

c. Accuracy of instrumentation used.

d. Descriptions and photographs of all test setups and a listing of equipment used.

4.1.2 Acceptance. Quality conformance inspections shall be performed under the surveillance of the Government inspector on lots submitted for acceptance under contract. The quality conformance tests shall consist of individual tests and sampling tests.

4.1.2.1 Acceptance inspections/tests. Each nozzle shall be subjected to the inspections and tests in Table VII.

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4.1.2.1.1 Sampling tests. One nozzle shall be selected at random by the Government inspector from each lot, or each 100 units whichever is less, which has passed the individual tests. That nozzle shall be subjected to the tests in Table VIII.

4.1.2.1.2 Lots. A lot shall consist of nozzles of the same part number manufactured under essentially the same conditions and submitted for inspection at the same time.

4.1.2.2 Acceptance failures.

4.1.2.2.1 Rejection. When one or more nozzles from a lot fail to meet the requirements of this specification, acceptance of all nozzles in the lot shall be withheld until the extent and cause of the failure are determined.

4.1.2.2.2 Resubmittal. Before resubmitting, full particulars concerning the failures and necessary corrective actions shall be furnished to the qualifying activity.

4.1.2.2.3 Retest. After corrective actions have been taken, all conformance inspections shall be repeated.

4.1.2.2.4 Resubmittal prohibited. Nozzles rejected after retest shall not be submitted for further testing without specific approval of the qualifying activity.

4.2 Test and inspections.

4.2.1 Test and inspection conditions. Unless otherwise specified, tests and inspections shall be performed at ambient atmospheric pressures and ambient air temperatures between 60 and 90 °F with the test fluid conforming to MIL-DTL-5624, MIL-DTL-83133, or MIL-PRF-7024.

4.2.2 Tolerances. Unless otherwise specified, test pressures and flow rates within the verification section shall have tolerances in accordance with Table VI.

4.2.3 Test couplings. Unless otherwise agreed to by the qualifying activity, couplings used in the following tests and evaluations shall be any coupling found in QPL MIL-PRF-81975 with a latching mechanism adjusted to produce a disengagement force of 420 ±20 pounds at 0 psig internal pressure. Test coupling shall successfully pass the Acceptance Test Procedure as defined in MIL-PRF-81975.

4.2.4 Test cycles. Unless otherwise specified, the engagement of the nozzle, its sealing in the coupling, the fuel pressurization of the nozzle while in the engaged/sealed condition with the coupling (when required), and the complete disengagement of the nozzle from the coupling shall constitute one complete cycle of operation.

4.3 Verification requirements.

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4.3.1 Assembly. The nozzle shall be inspected for conformance to the dimensions shown on Figure 1.

4.3.2 Materials.

4.3.2.1 Fluid resistance. All materials used in the construction of the nozzle shall be shown to be resistant to the fluids listed in 3.2.1, by testing in accordance with Table I.

4.3.2.2 Corrosion-resistant materials. The nozzle drawings shall be inspected for corrosion-resistant protective measures.

4.3.2.2.1 Protective coatings. After qualification testing the nozzle shall be inspected for chipping, cracking, or flaking protective coatings. Inspect nozzle drawings for compliance of cadmium plating to SAE-AMS-QQ-P-416 and that there are no painted surfaces.

4.3.2.2.2 Aluminum protection. The nozzle drawings shall be inspected to confirm aluminum alloy parts are protected in accordance with 3.2.2.2.

4.3.2.3 Prohibited materials. The nozzle materials list shall be inspected for exclusion of prohibited materials given in 3.2.3.

4.3.2.3.1 Impact materials. Analysis and inspection of nozzle drawings shall show that PTFE is not used in any location where impact loads may be encountered.

4.3.2.3.2 Age-sensitive materials. The nozzle drawings shall be inspected for exclusion of age-sensitive materials, except those specified in 3.4.1.

4.3.2.4 Castings. If present, the nozzle drawings, and production documents shall be inspected for compliance with 3.2.4.

4.3.2.5 Dissimilar metals. The nozzle assembly drawings shall be analyzed to verify compliance with 3.2.5.

4.3.2.6 Electrical protection.

4.3.2.6.1 Electrical bonding. The nozzle electrical bonding resistance capability shall be verified utilizing a representative probe mast per the following procedure:

1. Establish a low resistance contact point on the nozzle body. This may be achieved by mechanically removing the surface treatment locally from the nozzle body as shown on Figure 9, or by using an existing low resistance feature on the nozzle body.

2. Install nozzle onto probe mast per MS24354, using MS24360-2 lock ring (note: it is not necessary to deform the lock ring tab). The MIL-DTL-25988/1B O-ring may be omitted for ease of installation.

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3. Torque nozzle to between 494 and 543 in-lbs.

4. Using the 4-point probe method, measure the electrical bonding resistance between the contact point established in STEP 1 and the surface of the probe mast as shown on Figure 9.

5. The electrical bonding resistance shall not exceed 1.0 Ohm.

4.3.2.7 Synthetic rubber parts. All synthetic rubber parts shall be inspected for markings in accordance with 3.2.7.

4.3.3 Selection of specification and standards. Verify necessary commodities and services not specified herein are in accordance with applicable aerospace practices.

4.3.4 Construction. The nozzle shall be inspected to ensure compliance with 3.4.

4.3.4.1 O-rings, packings, and seals. The nozzle drawings, O-rings, packing, and seals shall be inspected for compliance with 3.4.1.

4.3.4.2 Threaded parts. The nozzle drawings and hardware screw threads shall be inspected for conformance to MIL-S-7742 or SAE AS8879, or approval by the qualifying activity.

4.3.4.2.1 Prohibited threads. The nozzle drawings shall be inspected for exclusion of pipe threads.

4.3.4.2.2 Locking of threaded parts. The nozzle drawings and hardware shall be inspected for conformance to approved positively locking methods in accordance with 3.4.2.2.

4.3.4.2.2.1 Safety wiring. For threaded parts utilizing safety wiring the nozzle drawings and hardware shall be inspected for conformance to NASM20995 and NASM33540.

4.3.4.2.2.2 Self-locking nuts. For threaded parts utilizing self-locking nuts the nozzle drawings and hardware, shall be inspected for conformance to NASM25027 and NASM33588.

4.3.4.2.2.3 Prohibited locking. The nozzle drawings and hardware shall be inspected for exclusion of locking methods identified in 3.4.2.2.3.

4.3.4.3 Weight. The nozzle shall be weighed to demonstrate a total dry weight less than that given in 3.4.3.

4.3.4.4 Finishes. The nozzle finish shall be inspected for smoothness and exclusion of sharp edges.

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4.3.4.4.1 Surface roughness. The nozzle finish shall be inspected for conformance to Figure 1 in accordance with ASME B46.1.

4.3.4.5 Reliability. This requirement is verified by successful completion of all environmental verifications in 4.3.5.14 and endurance verifications in 4.3.5.14.4.2.

4.3.4.6 Maintainability. Verify no maintenance is required throughout all qualification testing as defined in Table IV and Table V.

4.3.4.7 Interchangeability. The nozzle drawings shall be inspected to verify part interchangeability in accordance with 3.4.7.

4.3.4.8 Dimensions. The nozzle hardware shall be inspected for conformance to the dimensions shown on Figure 1.

4.3.4.9 Lubrication. The nozzle, drawings, and qualification test reports shall be inspected to verify that lubrication, other than the fluid being transferred, is not required for the nozzle to function.

4.3.4.10 Identification of product. The nozzle equipment, assemblies, and parts shall be inspected for identification markings conforming to MIL-STD-130 or MIL-STD-190 as applicable.

4.3.4.10.1 Application. The nozzle equipment, assemblies, and parts identification markings shall be inspected at the completion of qualification testing to ensure there is no peeling, cracking, or loosening of the marking.

4.3.4.10.2 Location. The nozzle and drawings shall be inspected to verify the markings do not affect the nozzle performance.

4.3.4.10.3 Markings. The nozzle and drawings shall be inspected for the inclusion of the information identified in 3.4.10.3.

4.3.4.10.4 Part number. The nozzle and drawings shall be inspected to verify the part numbers conform to ASME Y14.100 identification and part number requirements.

4.3.4.11 Workmanship. The nozzle shall be inspected for uniformity, cleanliness, lack of defects, proper alignment, and removal of burrs.

4.3.4.12 Incorrect assembly. The nozzle drawings shall be inspected for potential of incorrect assembly.

4.3.5 Operation.

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4.3.5.1 Fuel transfer. The nozzle when engaged with a test coupling shall be tested to verify fuel transfer.

4.3.5.2 Installation. The nozzle to mast interface shall be verified by inspection of the supplier's MA-2 nozzle drawing for conformance with Figure 7, and by physical installation on a mating probe mast conforming with Figure 7.

4.3.5.2.1 Nozzle sleeve operation. The nozzle shall be tested to verify that the sleeve can't be retracted until the sleeve latches are depressed flush with the nozzle body surface.

4.3.5.3 Latching mechanism. The nozzle and drawings shall be inspected for the inclusion of exactly two sleeve latches spaced 180 degrees apart.

4.3.5.3.1 Engagement. A sleeve latch test gauge, as shown on Figure 8, shall be passed along the nozzle's longitudinal axis in the engaging direction while the sleeve is closed. There shall be no evidence of binding or failure. The force required to depress the nozzle sleeve latches shall be as specified in 3.5.3.1.

4.3.5.3.2 Disengagement. A sleeve latch test gauge, as shown on Figure 8, shall be passed along the nozzle's longitudinal axis in the disengaging direction while the sleeve is closed. There shall be no evidence of binding or failure. The force required to depress the nozzle sleeve latches shall be as specified in 3.5.3.2.

4.3.5.4 External power source. The nozzle and drawings shall be inspected to ensure that no external power is required for operation.

4.3.5.5 Off-center disconnects. The nozzle shall be disengaged 5 times at an angle of 22.5 degrees. Disconnect forces shall be applied at the coupling hose connection interface per Figure 6. A compliant connection to the hose coupling interface may be used to allow flexure of the system during disconnect. There shall be no evidence of binding or failure.

4.3.5.6 Emergency disconnect. While maintaining 50 ±2 psig fuel pressure in the coupling, the nozzle shall be disengaged 5 times at an angle of 22.5 degrees from the nozzle longitudinal axis. Disconnect forces shall be applied at the coupling hose connection interface per Figure 6. A compliant connection to the hose coupling interface may be used to allow flexure of the system during disconnect. No evidence of binding or failure shall occur.

4.3.5.7 Proof pressure.

4.3.5.7.1 Engaged proof pressure. With the nozzle engaged with a coupling and the coupling inlet blanked off, the nozzle shall be subjected to a hydrostatic pressure of 240 ±2 psig for a minimum of one minute. There shall be no evidence of external leakage, permanent deformation, structural failure, binding of the nozzle sleeve, or other injury to any part of the nozzle throughout the required test period.

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4.3.5.7.2 Disengaged proof pressure. The nozzle alone shall be subjected to a hydrostatic pressure of 240 ±2 psig for a minimum of one minute. There shall be no evidence of external leakage, deformation, or other injury to any part of the nozzle throughout the required test period.

4.3.5.8 Ultimate (burst) pressure.

4.3.5.8.1 Engaged ultimate pressure. With the nozzle engaged with a coupling and the coupling inlet blanked off, the nozzle shall be subjected to an internal hydrostatic pressure of 360 ±2 psig for a minimum of one minute. It shall meet the requirements of 3.5.8.1.

4.3.5.8.2 Disengaged ultimate pressure. The nozzle alone shall be subjected to an internal hydrostatic pressure of 360 ±2 psig for a minimum of one minute. It shall meet the requirements of 3.5.8.2. The pressure shall then be lowered to 60 psig. There shall be no evidence of external leakage.

4.3.5.9 Leakage and spillage. The attitude of the coupling shall be most conducive for the drainage of leakage which may be retained internally by the coupling and nozzle for the following tests.

4.3.5.9.1 Initial allowances. This test shall be performed as follows:

a. The engaged nozzle and coupling shall be subjected to constant inlet pressures of 2 ±0.5 and 60 ±2 psig for a minimum of 6 minutes each. There shall be no external leakage at each pressure in excess of the limit specified in 3.5.9.1(a).

b. The engaged nozzle and coupling shall be subjected to an inlet fluid pressure of 50 -5/+0 psig and a fluid flow of 100 to 600 – 0/+30 gal/min in increments of 100 gal/min. The nozzle shall be disengaged at every increment at 5 ft/s minimum velocity. The minimum velocity shall be achieved within 2 ±1 inches of travel. The spillage per disengagement shall not be greater than the limit specified in 3.5.9.1(b).

c. The coupling shall be subjected to an inlet fluid pressure of 10 ±1 psig. The nozzle shall be engaged and disengaged a minimum of five times. Spillage per engagement or disengagement shall not be greater than the limit specified in 3.5.9.1(c).

4.3.5.9.2 Subsequent allowances. This test shall be performed as follows:

a. The engaged nozzle and coupling shall be subjected to constant inlet pressures of 2 ±0.5 and 60 ±2 psig for a minimum of 6 minutes each. There shall be no seepage at each pressure in excess of the limit specified in 3.5.9.2(a).

b. The engaged nozzle and coupling shall be subjected to an inlet fluid pressure of 50 -5/+0 psig and a fluid flow of 100 to 600 – 0/+30 gal/min in increments of 100 gal/min. The nozzle shall be disengaged at every increment at 5 ft/s minimum velocity. The minimum velocity shall be achieved within 2 ±1 inches of travel. The spillage per disengagement shall

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not be greater than the limit specified in 3.5.9.2(b).

c. The coupling shall be subjected to an inlet fluid pressure of 10 ±1 psig. The nozzle shall be engaged and disengaged a minimum of five times. Spillage per engagement or disengagement shall not be greater than the limit specified in 3.5.9.2(c).

4.3.5.9.3 Immersion leakage. The nozzle shall be completely submerged in water for 30 minutes with the sleeve closed. A negative pressure of 2 psid (4 inHg), below ambient, shall be maintained within the nozzle. There shall be no leakage of water into the nozzle.

4.3.5.10 Pressure drop. The test shall be conducted with the nozzle engaged with a reception coupling, all regulators inactive, and a pipe inner diameter of 2.875 inches ±0.1 inch. The pressure drop shall be measured from the coupling latch port to the nozzle outlet at 50 gpm increments over the flow rate range of 0 to 600 -0/+30 gpm (for report only) with an inlet static pressure of 50 ±2 psig. At 600 -0/+30 gpm the pressure drop shall not be greater than the value in section 3.5.10. If test fluid meeting MIL-PRF-7024 is used, the following pressure drop correction formula shall be used:

ΔPCORR = ΔPREC * [1.34/VISCOSITY OF TEST FLUID (CS)]0.25 * [0.770/SG OF TEST FLUID]

Where ΔPREC is the actual recorded pressure drop from the test, CS is centistokes, and SG is Specific Gravity.

4.3.5.11 Contaminated fuel. The test fluid shall contain the types and concentrations of contaminants specified in Table II. The engaged nozzle and coupling shall be subjected to a flow rate of at least 50 gpm and operated for 1,000 cycles (a cycle shall consist of engaging for 30 seconds and disengaging). The flow-generating device shall ensure that the contaminants remain in uniform suspension in the fluid. Systems that utilize recirculated contaminants shall be designed such that contaminants are continuously recirculated back to the flow-generating device without falling out of suspension in the fluid. Upon completion, the nozzle shall be flushed and drained and pass the functional tests described in 4.3.5.17.

4.3.5.12 Design limit loads.

4.3.5.12.1 Limit loads. The nozzle shall be mounted upon a rigid mast (see Figure 6). The nozzle shall then be subjected to the loads specified in 3.5.12.1 applied as shown on Figure 4. Tensile loads shall be applied at the latching shoulder parallel to the longitudinal axis of the nozzle, and compression loads shall be applied at the lip of the nozzle sleeve parallel to the longitudinal axis of the nozzle. Radial loads shall be applied to the nozzle sleeve 3.5 inches from the gage point in the roller locking groove perpendicular to the longitudinal axis of the nozzle. Compression loads shall be applied with the nozzle sleeve in both the open and closed position. The nozzle shall withstand the applied loads without permanent deformation, structural failure, or any other failure that would render it inoperable.

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4.3.5.12.1.1 Follow-on tests. Immediately after the test in 4.3.5.12.1 the nozzle shall pass the functional tests of 4.3.5.17 and the proof pressure test of 4.3.5.7.2 (subsequent).

4.3.5.12.2 Nozzle tip deflection. The nozzle shall be installed on a test fixture. The nozzle sleeve shall be held in the fully open position while the nozzle tip is deflected radially until there is contact with the sleeve lip. With the nozzle sleeve held in the fully open position, the nozzle tip shall then be released and shall return to its original, un-deflected position within 1/16-inch.

4.3.5.12.2.1 Follow-on tests. Immediately after the test in 4.3.5.12.2, the nozzle sleeve shall be released to its closed (spring loaded position) and the nozzle shall pass the pressure test section of 4.3.5.17.1.

4.3.5.13 Nozzle-coupling compatibility. This test shall be performed using a fuel reception test coupling as defined in 4.2.3, or as required by the procuring activity. The nozzle shall be engaged with a coupling so that the nozzle's sleeve latches are aligned with the coupling's locking roller windows in a manner which will produce the highest likelihood of binding or hang-up during engagement and disengagement. The force to engage the nozzle shall be measured. The nozzle shall then be disengaged, and the disengagement force measured. No binding or hang-up shall occur. The disengagement force shall be within ±30 pounds of the coupling latch-force defined in 4.2.3. The nozzle and the coupling shall then be re-engaged. With the nozzle held fixed, the coupling shall be rotated by the outlet flange about its longitudinal axis and about the fixed nozzle through 360 degrees in both directions. No hang-ups shall occur.

4.3.5.14 Environmental requirements.

4.3.5.14.1 Salt fog. The nozzle shall be installed in a test chamber in such a way that the sleeve remains closed, and the threaded end shall be sealed against moisture. A salt solution of 5 ±1 percent sodium chloride in distilled water shall be atomized within the test chamber. At least 0.3 quart (US) of salt solution shall be atomized per cubic foot of test chamber volume per 24 hours. Fallout rate of the salt fog shall be measured at least once every 24 hours and shall be between 0.1 and 0.61 teaspoon (US)/hour per 12.4 in2 of collection area. Test chamber temperature and salt solution temperature prior to atomization shall be maintained at 95 ±5 °F. The nozzle shall be exposed to salt fog for 48 hours, after which it shall be dried in ambient air as specified in 4.2.1 for 48 hours. The nozzle shall then be flushed with warm water, dried, and wetted with test fluid. Post Salt Fog test requirements are listed in Table V. Corrosion to an extent which could cause malfunction of the nozzle or contamination of the aircraft fuel system shall be cause for rejection.

4.3.5.14.2 Dust. The nozzle shall be subjected to blown dust in the manner specified in 4.3.5.14.2.1 to 4.3.5.14.2.2. At the conclusion of the test the remaining loose dust in or on the nozzle shall be blown away. Post Dust test requirements are listed in Table V.

4.3.5.14.2.1 Test conditions.

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4.3.5.14.2.1.1 Dust composition. The dust used shall consist of silica flour or red China clay with the following size distribution:

100 percent shall pass through a 100-mesh screen.
98 ±2 percent shall pass through a 140-mesh screen.
90 ±2 percent shall pass through a 200-mesh screen.
75 ±2 percent shall pass through a 325-mesh screen.

4.3.5.14.2.1.2 Air velocity. For the sand test, the air velocity shall be maintained between 3,450 and 5,700 feet per minute (fpm). For the dust test, the air velocity shall be maintained between 300 and 500 fpm.

4.3.5.14.2.1.3 Dust concentrations. Dust concentration shall be maintained at 4.632 grain/ft3.

4.3.5.14.2.2 Test procedure. The nozzle's outlet shall be sealed, and the nozzle shall be mounted in such a way as to allow dust to be blown upon the nozzle from different directions. Each test shall be run two times: once with the nozzle's tip pointed directly into the flow, and once with the nozzle's axis aligned perpendicularly to the flow. The nozzle shall be subjected to blowing dust for six hours at room temperature and six hours at an ambient air temperature of 185 ±10 °F. The nozzle's temperature shall be allowed to stabilize at the ambient air temperature prior to each test run.

4.3.5.14.3 Vibration. The vibration test shall be performed in accordance with MIL-STD-810H, Test Method 514.8 Procedure IV and Figure 5. This test shall be performed with the nozzle in a dry condition and mounted to the test fixture horizontally to simulate configuration of nozzle installed on aircraft. For the purpose of this test, vibration shall be applied along the direction of the major axis and two perpendicular minor axes, all of which are mutually perpendicular, for one hour each. Post vibration test requirements are listed in Table V. Additionally, the nozzle shall be inspected for structural failure; cracking, breaking, or loosening of parts; or permanent deformation; the occurrence of which shall constitute failure.

4.3.5.14.4 Impact and endurance.

4.3.5.14.4.1 Impact. The nozzle shall be installed upon the fixture shown on Figure 6 or other setup approved by the qualifying activity and subjected to the impact conditions described in Table III. Unless otherwise specified, an internal fluid pressure of 10 ±2 psig shall be maintained in the coupling throughout this test. Post Impact test requirements are listed in Table V.

4.3.5.14.4.2 Endurance. The nozzle shall be installed upon the fixture shown on Figure 6 or other setup approved by the qualifying activity and subjected to the endurance conditions described in Table III. Unless otherwise specified, an internal fluid pressure of 10 ±2 psig shall be maintained in the coupling throughout this test. Post Endurance test

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requirements are listed in Table V.

4.3.5.15 Extreme fuel and ambient temperature. The nozzle shall be subjected to tests and follow-on tests in accordance with Table I.

4.3.5.16 Disassembly and inspection. Upon completion of testing, the nozzle shall be disassembled, and all parts inspected. Parts which are distorted, weakened, or demonstrate abnormal wear, shall constitute failure.

4.3.5.17 Functional.

4.3.5.17.1 Disengaged leakage. The nozzle shall be checked at hydrostatic pressures of 2 ±0.5 psig and 60 ±2 psig for one minute at each pressure. The nozzle shall meet the requirements of 3.5.17.1. While the nozzle is pressurized to 60 psig, an opening force of 50 pounds shall be applied to the lip of the sleeve without depressing the sleeve latches. No leakage shall occur. This test shall then be repeated using an internal pressure of 2 psig. No leakage shall occur.

4.3.5.17.2 Engaged leakage. The nozzle, engaged with a coupling, shall be pressurized to 2 ±0.5 psig and 60 ±2 psig for one minute at each pressure. The nozzle shall meet the requirements of 3.5.17.2.

4.3.5.17.3 Opening sleeve force. The nozzle sleeve shall be opened, with the sleeve latches depressed, to its fully open position. A record of force applied (lbf) versus sleeve movement (in.) shall be obtained. A minimum of six data points shall be recorded. The force required to fully open the sleeve shall be measured and meet the requirements of 3.5.17.3.

4.3.5.17.4 Closing sleeve force. With the nozzle sleeve full open, the force applied shall be gradually reduced until the sleeve begins to close. This force shall be recorded. The force shall be gradually further reduced until the sleeve seals shut. The force against which the sleeve seals shall not be less than 15 pounds as specified in 3.5.17.4. There shall be no evidence of binding or chattering during sleeve movement.

4.3.5.17.5 Engagement and disengagement. The coupling utilized during this testing shall have its latching mechanism adjusted to produce a disengaging force of 420 ±20 pounds at zero psig fuel pressure. The following tests shall then be conducted:

a. With the nozzle engaged in the coupling and while maintaining a 10 ±1 psig fuel pressure in the coupling, the nozzle shall be disengaged and engaged at least five times (nozzle does not need to be held at 10 ±1 psig fuel pressure while disengaged from coupling). No evidence of binding, chattering, or excessive forces shall occur as specified in 3.5.17.5.

5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract or order (see 6.2), the supplier shall package and protect all parts in a manner that will prevent damage and

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deterioration during shipment and storage. The supplier shall not insert packaging data into the nozzle. When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Attachment Parts. Unless otherwise instructed in the contract, or other ordering instrument, the attachment parts required for proper installation of the probe nozzle, 1 each MS24360-2 Lock Ring and 1 each M25988/1-234 O-ring, shall be included in the unit packaging with the probe nozzle. These parts shall be packaged with the parts installed in their proper relative position on the probe nozzle, or otherwise secured to the probe nozzle in a manner to prevent loss during storage, handling, and installation.

5.3 Marking of shipments. Exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Instructions. The following instruction shall be marked on the exterior container for each individual unit:

IF IN STORAGE FOR 20 YEARS OR LONGER, THE NOZZLE SHALL BE PROOF PRESSURE TESTED AND INSPECTED BEFORE USE.

6. NOTES

6.1 Intended use. The aerial refueling probe nozzles covered by this specification are intended for use by aircraft being refueled in flight from tanker aircraft by means of the probe-drogue method of aerial refueling. The nozzle, which is installed on the receiver aircraft being refueled, engages the aerial refueling reception coupling extended from the tanker aircraft, thereby allowing the transfer of fuel. The nozzle may also be used for single point ground refueling with the aid of a special adapter.

6.2 Acquisition requirements. Acquisition documents will specify the following:

a. Title, number, and date of this specification.

b. Levels of preservation, packaging, and packing. (see Section 5)

c. Lot inspections/tests. One nozzle selected by the inspector from each lot of 100 or fraction thereof that has passed the individual tests will be subjected to the tests in Table VIII.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List, QPL-25161 whether or not such products have actually been listed by

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that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Department of the USAF. An online listing of products qualified to this specification are located in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Definitions.

6.4.1 Abnormal wear. Any unusual, excessive, or premature wear that would potentially cause the unit to function improperly.

6.4.2 Age-sensitive material. An age-sensitive material is one whose physical properties or form slowly degrades beyond the desired performance range due to exposure to typical environmental conditions. These conditions include, but are not limited to, oxygen, ozone, (especially in the presence of ultraviolet light), moisture, high temperatures, swelling/leaching agents (fuel and solvents), corrosive vapors, and mechanical stress.

6.4.3 Continuous leakage. When a fuel leak from the nozzle is still evident 10 seconds after the initiation of disconnection.

6.4.4 CRES. Corrosion resistant steel.

6.4.5 Dissimilar metals. The occurrence of dissimilar metals is created when two metal specimens are in contact, or otherwise electrically connected to each other in a conductive solution, and an electric current is capable of being generated between the two metal specimens. Dissimilar metals are defined in MIL-STD-889.

6.4.6 Engaged. The position of the nozzle when completely seated in the reception coupling. In the engaged position, the three coupling locking rollers are fully seated in the nozzle roller locking groove. The roller locking groove is identified on Figure 3.

6.4.7 External leakage. Wetting sufficient to form a drop.

6.4.8 Hydrostatic pressure. The pressure of a fluid with no dynamic effects; usually measured with a port perpendicular to the fluid flow.

6.4.9 Interchangeable. Parts that are interchangeable are manufactured with the aid of controlled media and require only the application of attaching means for their installation. These parts are capable of being readily installed, removed, and replaced without alteration, misalignment, or damage to themselves or to the adjoining items or structure.

6.4.10 NASM. National Aerospace Standard, Military (English Units).

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6.4.11 Psid. Pounds per square inch differential; difference in pressure between two points.

6.4.12 Psig. Pounds per square inch gauge; pressure relative to atmospheric pressure.

6.4.13 Qualifying activity. The engineering home office of the specification's primary custodian.

6.5 International standardization agreement compliance. This specification complies with STANAG 3447 (ATP 3.3.4.6), Air-to-Air (Aerial) Refueling Equipment: Probe-Drogue Interface Characteristics. When amendment, revision, or cancellation of this specification is proposed, the preparing activity will coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.6 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TABLE I. Fluid resistance and extreme temperature.

Test	Fuel Resistance				Low Temperature
Period[1]	Phase I soak[2]	Phase I dry	Phase II soak[2]	Phase II dry	Phase III soak[2]
Nozzle Configuration		Drained and blown dry		Drained and blown dry	
Test fluid	ASTM D5900, IRM 903[3]	None	ASTM D5900, IRM 903[3]	None	ASTM D5900, IRM 901
Minimum period duration	96 hours	24 hours	18 hours	30 hours	18 hours
Ambient and test fluid temperature		Circulating air at:		Circulating air at:	
Class A[4]	158 ±2 °F	158 ±2 °F	158 ±2 °F	158 ±2 °F	-67 ±2 °F
Class B[4]	236 ±5 °F	350 ±5 °F	236 ±5 °F	350 ±5 °F	-67 ±2 °F
Operation and tests during period	Actuate nozzle sleeve at least four times per day. The sleeve shall not hang-up.	None	Actuate nozzle sleeve at least four times per day. The sleeve shall not hang-up.	None	None

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TABLE I. Fluid resistance and extreme temperature – Continued.

Test	Fuel Resistance				Low Temperature
Operation and tests immediately after period	Perform functional tests using ASTM D5900, IRM 903.	Actuate nozzle sleeve at least four times. There shall be no sleeve hang-up. Perform functional test using ASTM D5900, IRM 901.	Perform functional tests using T ASTM D5900, IRM 903.	Actuate nozzle sleeve at least four times. There shall be no sleeve hang-up. Perform functional test using ASTM D5900, IRM 901.	Maintaining the ambient temperature and test fluid at -67 ±2°F the nozzle shall be subjected to fuel pressures of 2 and 60 psig for periods of 15 minutes. There shall be no leakage. The pressure shall then be relieved, and the sleeve actuated four times. After each sleeve closure a pressure of 2 psig shall be applied to assure proper sleeve sealing. There shall be no sleeve hang-up nor leakage.

Notes:
1. Each period shall follow immediately after the preceding one in the order noted.
2. The nozzle shall be subjected to the test fluid in such a manner to assure complete contact, as would be expected under normal service conditions.
3. For class B application, use MIL-PRF-680 at sufficient pressure (less than 15 psig) to prevent boiling.
4. Class for qualifications is determined by operating range expected in normal service use. (See 1.2.2)

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TABLE II. Contaminant mixture.

Contaminant	Particle Size (Microns)	Quantity (grams per 1000 liters)
Iron oxide	0 – 5 5 - 10	19.00 1.00
Sharp silica sand	150 – 300 300 - 420	0.70 0.70
Prepared dirt conforming to AC Spark Plug Co. Part No. 1543637 (Coarse Arizona road dust), or equivalent	Mixture as follows: 0 - 5 (12 percent) 5 - 10 (12 percent) 10 - 20 (14 percent) 20 - 40 (23 percent) 40 - 80 (30 percent) 80 - 200 (9 percent)	5.30
Cotton linters	Staple below 7 US Dept of Agriculture Grading Standards	0.07
Iron Chips Aluminum Chips	150-500 150-500	10.00 10.00
Salt water solution (5 ±1 percent by weight NaCl)		0.01 percent entrained

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TABLE III. Impact and endurance test.

Condition[1]	Displacement of coupling centerline from nozzle centerline (inches)		Drop distance nozzle tip to coupling poppet (inches)	Angle of disconnect (degrees)	Initial, dry latch-force setting[7] (lb ±20)	Cycles[2]
Impact[3]	0 2 4		18 18 20	0 0 0	500 500 500	100[4] 400 500
Endurance[5]	0		9 9	22.5 0	500 500	1250[4] 2750[4, 8]

NOTES:

1. Thirty percent of the cycles at each condition shall be performed at -67 ±2 °F.
2. A cycle is defined as one engagement and one disengagement.
3. During the impact test, engagement shall be complete. If impact does not cause engagement, manual force shall be applied as required to complete engagement.
4. Fifty of the specified cycles shall be accomplished with zero fuel pressure and a dry nozzle.
5. Drop test shall not be required. Mechanical engagement shall be permissible, provided engaging velocity is not less than 5 fps just prior to seating the nozzle sleeve upon the master coupling seal.
6. Disengagement shall be accomplished with 50 ±2 psig fuel pressure applied to the reception coupling.
7. The term “Latch Force Setting” refers to the overall force required to disengage a probe from the coupling, also referred to as the “Disengagement force”.
8. The 1,000 cycles called out for the contaminated fuel test (see 4.3.5.11) may be considered as part of this test, provided the engaging velocity during the contaminated fuel test is not less than 5 fps.

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TABLE IV. Qualification test sequence #1.

Verification	Paragraph
Inspections	4.3.1 4.3.2.2 – 4.3.2.5 4.3.2.7 – 4.3.3 4.3.4.7 4.3.4.9
Weight	4.3.4.3
Dimensions	4.3.4.8
Functional	4.3.5.17
Leakage and Spillage (Initial)	4.3.5.9
Proof Pressure	4.3.5.7
*Leakage and Spillage (Initial)	4.3.5.9
*Functional	4.3.5.17
*Electrical Bonding	4.3.2.6
Pressure Drop	4.3.5.10
Fluid Resistance & Extreme Temperature	4.3.2.1 4.3.5.15
*Leakage and Spillage (Subsequent)	4.3.5.9
*Functional	4.3.5.17
*Electrical Bonding	4.3.2.6
Contaminated Fuel	4.3.5.11
*Leakage and Spillage (Subsequent)	4.3.5.9
*Functional	4.3.5.17
*Electrical Bonding	4.3.2.6
Latching Mechanism	4.3.5.3
Ultimate (Burst) Pressure	4.3.5.8
Disassembly and Inspection	4.3.5.16

* Indicates a required follow-on test

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TABLE V. Qualification test sequence #2.

Verification	Paragraph
Inspections	4.3.1 4.3.2.2 – 4.3.2.5 4.3.2.7 – 4.3.3 4.3.4.7 4.3.4.9
Weight	4.3.4.3
Dimensions	4.3.4.8
Functional	4.3.5.17
Electrical Bonding	4.3.2.6
Leakage and Spillage (Initial)	4.3.5.9
Nozzle-Coupling Compatibility	4.3.5.13
Vibration	4.3.5.14.3
*Leakage and Spillage (Subsequent)	4.3.5.9
*Functional	4.3.5.17
*Electrical Bonding	4.3.2.6
Salt Fog	4.3.5.14.1
*Leakage and Spillage (Subsequent)	4.3.5.9
*Functional	4.3.5.17
*Electrical Bonding	4.3.2.6
Dust	4.3.5.14.2
*Leakage and Spillage (Subsequent)	4.3.5.9
*Functional	4.3.5.17
*Electrical Bonding	4.3.2.6
Impact and Endurance	4.3.5.14.4
*Leakage and Spillage (Subsequent)	4.3.5.9
*Functional	4.3.5.17
Design Limit Loads	4.3.5.12
*Leakage and Spillage (Subsequent)	4.3.5.9
*Functional	4.3.5.17
*Electrical Bonding	4.3.2.6
Disassembly and Inspection	4.3.5.16

* Indicates a required follow-on test

TABLE VI. Verification tolerances.

Flow Rate	Tolerance		Pressure	Tolerance
≤ 1 gpm	± 5%		< 20 psi	± 10%
> 1 gpm	± 2%		≥ 20 psi	± 2 psi

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TABLE VII. Acceptance test sequence.

Verification	Paragraph
Inspections	4.3.1 4.3.2.2 – 4.3.2.5 4.3.2.7 – 4.3.3 4.3.4.7 4.3.4.9 4.3.5.3.1, 4.3.5.3.2
Weight	4.3.4.3
Dimensions	4.3.4.8
Proof Pressure	4.3.5.7
Functional	4.3.5.17
Leakage and Spillage (initial)	4.3.5.9.1a, 4.3.5.9.1c

TABLE VIII. Lot Sample test sequence.

Verification	Paragraph
Latching Mechanism	4.3.5.3.1, 4.3.5.3.2
Leakage and Spillage (Initial)	4.3.5.9.1b
Design Limit Loads	4.3.5.12
Emergency Disconnect	4.3.5.6
Leakage and Spillage (subsequent)	4.3.5.9.2
Electrical Bonding	4.3.2.6
Ultimate (Burst) Pressure	4.3.5.8
Disassembly and Inspection	4.3.5.16

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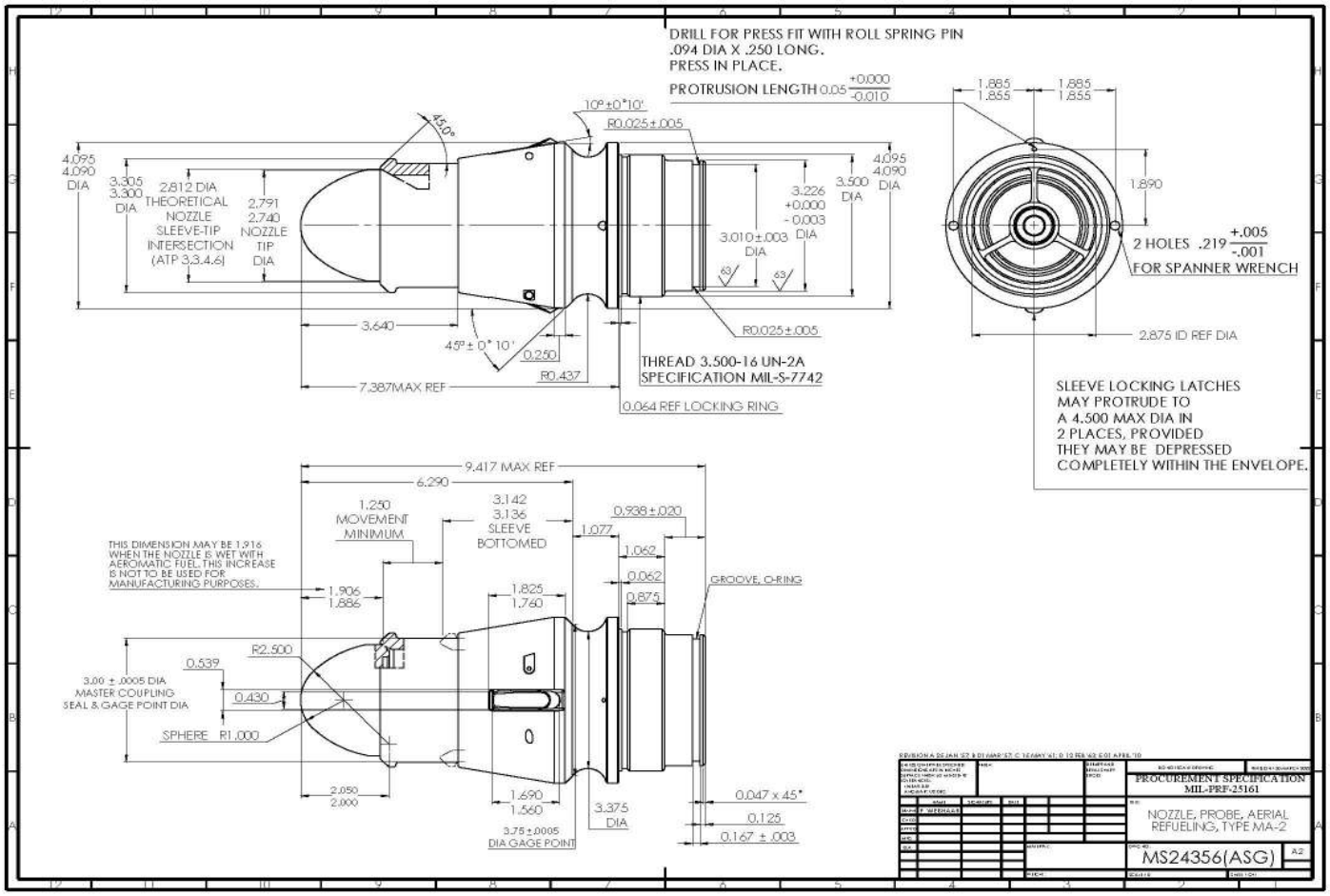
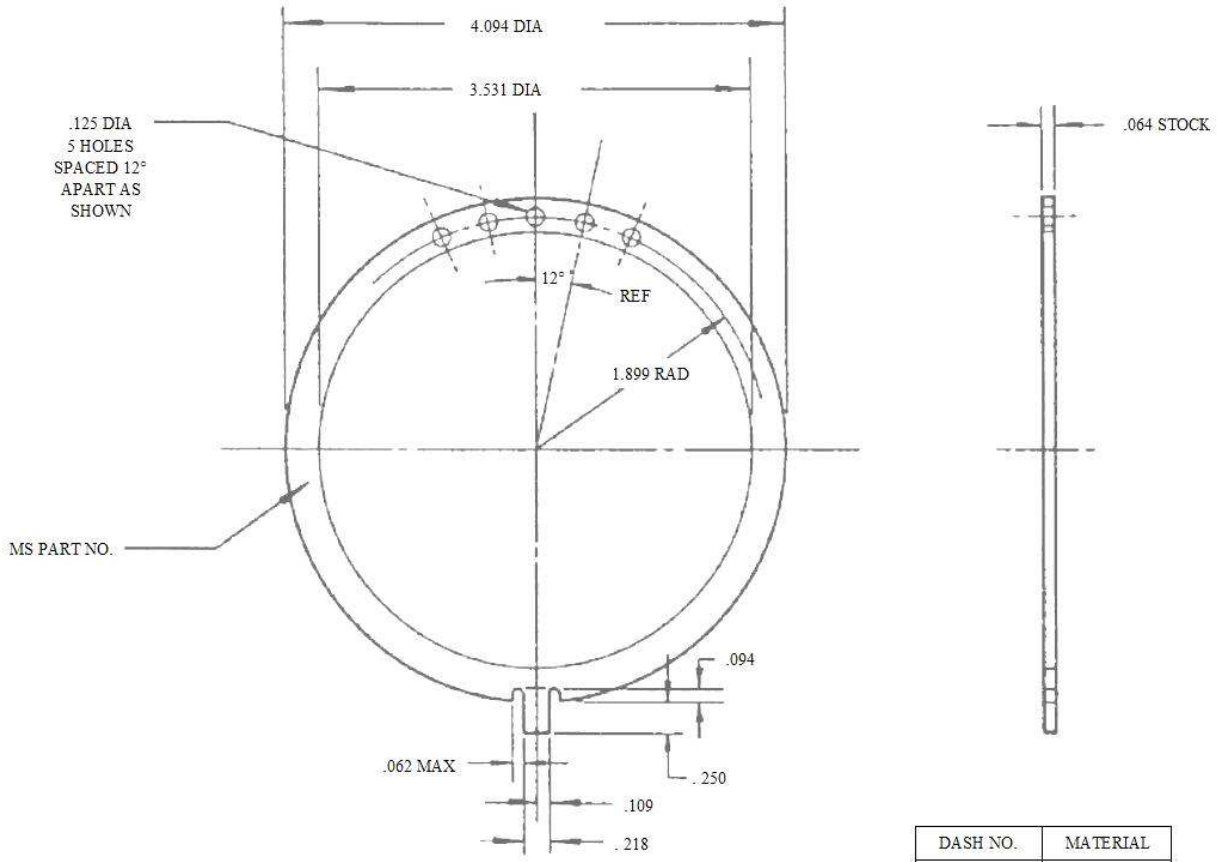


FIGURE 1. Nozzle dimensions.

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DASH NO.	MATERIAL
-1	ALUMINUM
-2	CRES

FIGURE 2. Lock ring configuration and dimensions.

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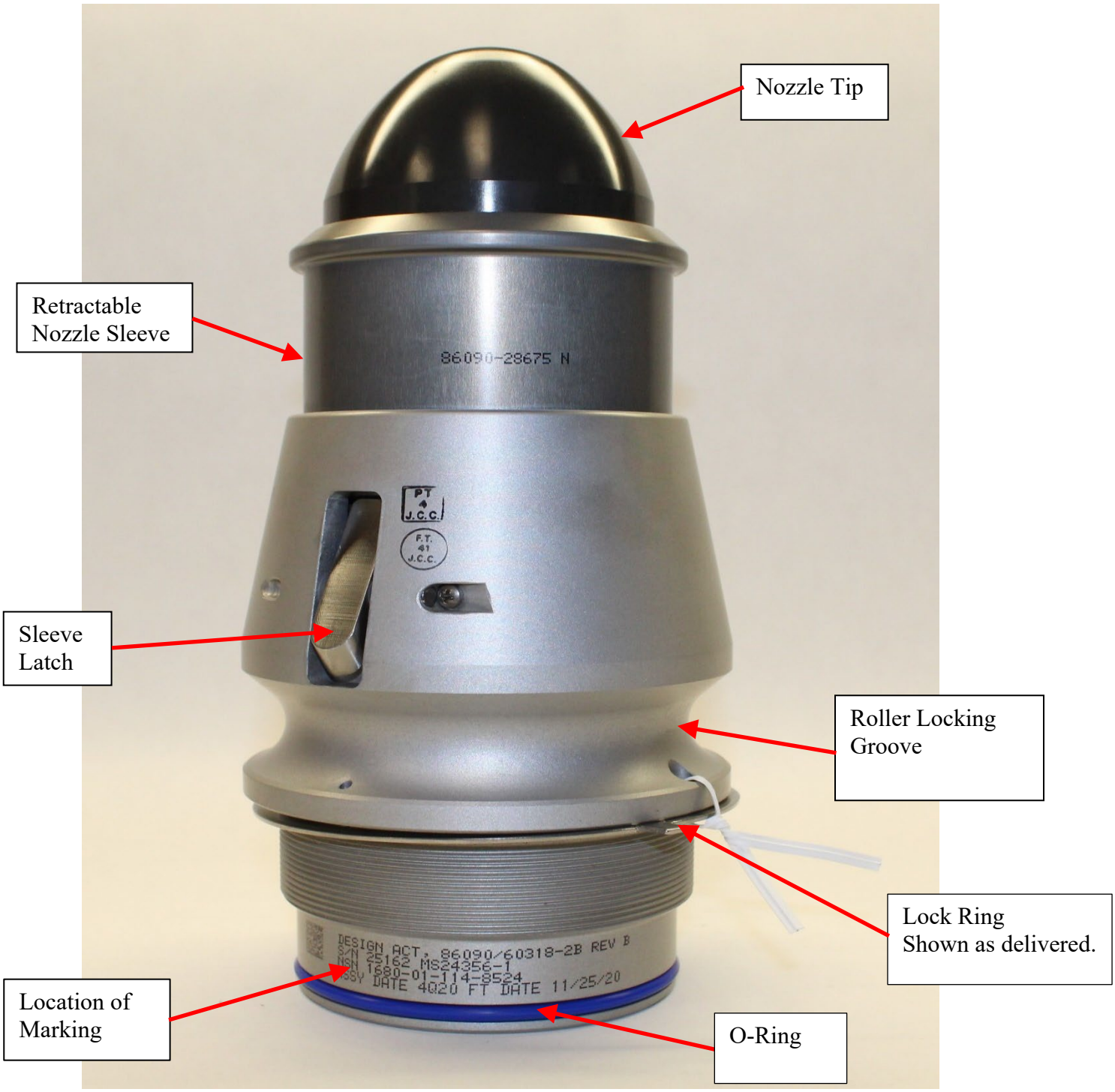


FIGURE 3. Nozzle components.

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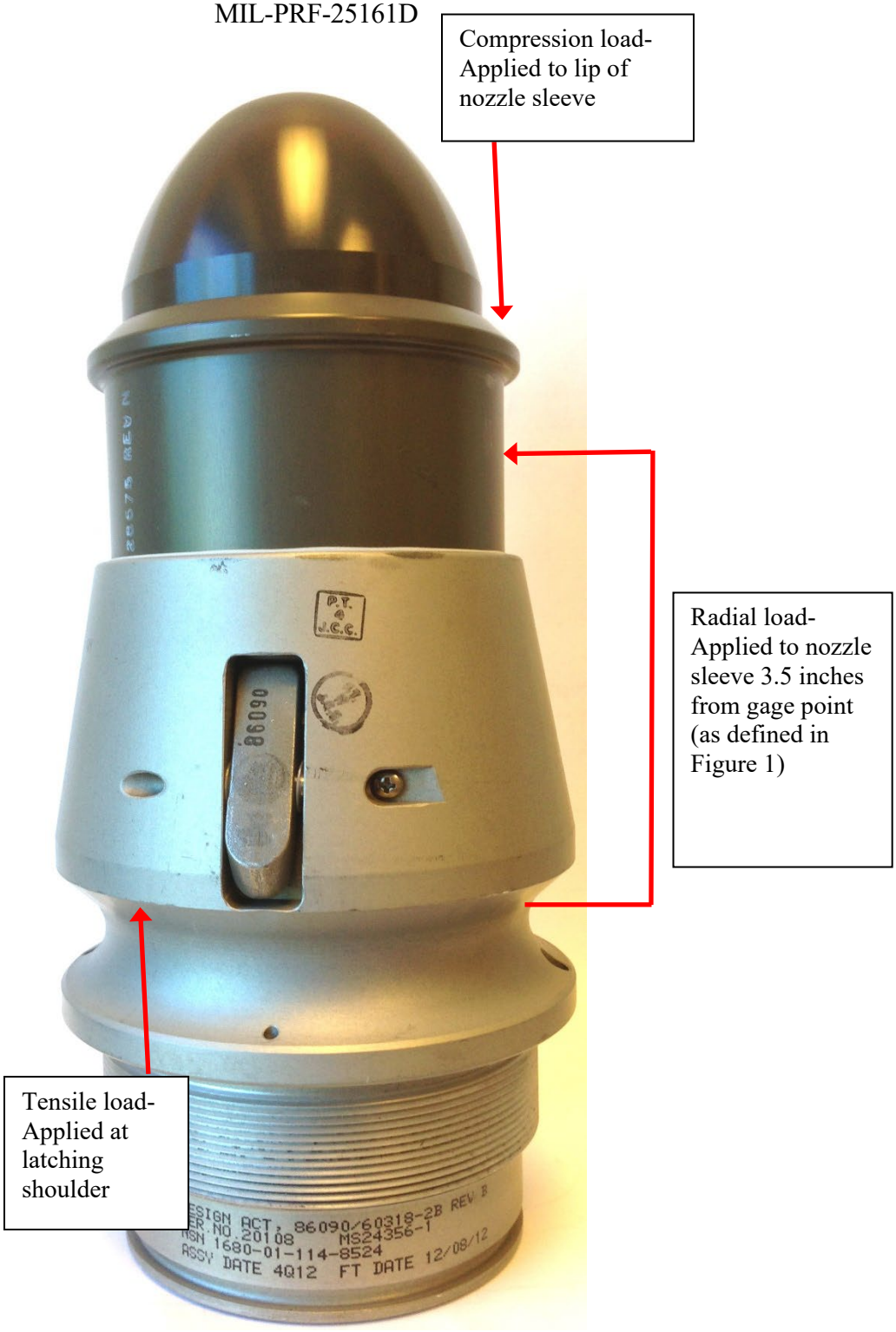


FIGURE 4. Load application points.

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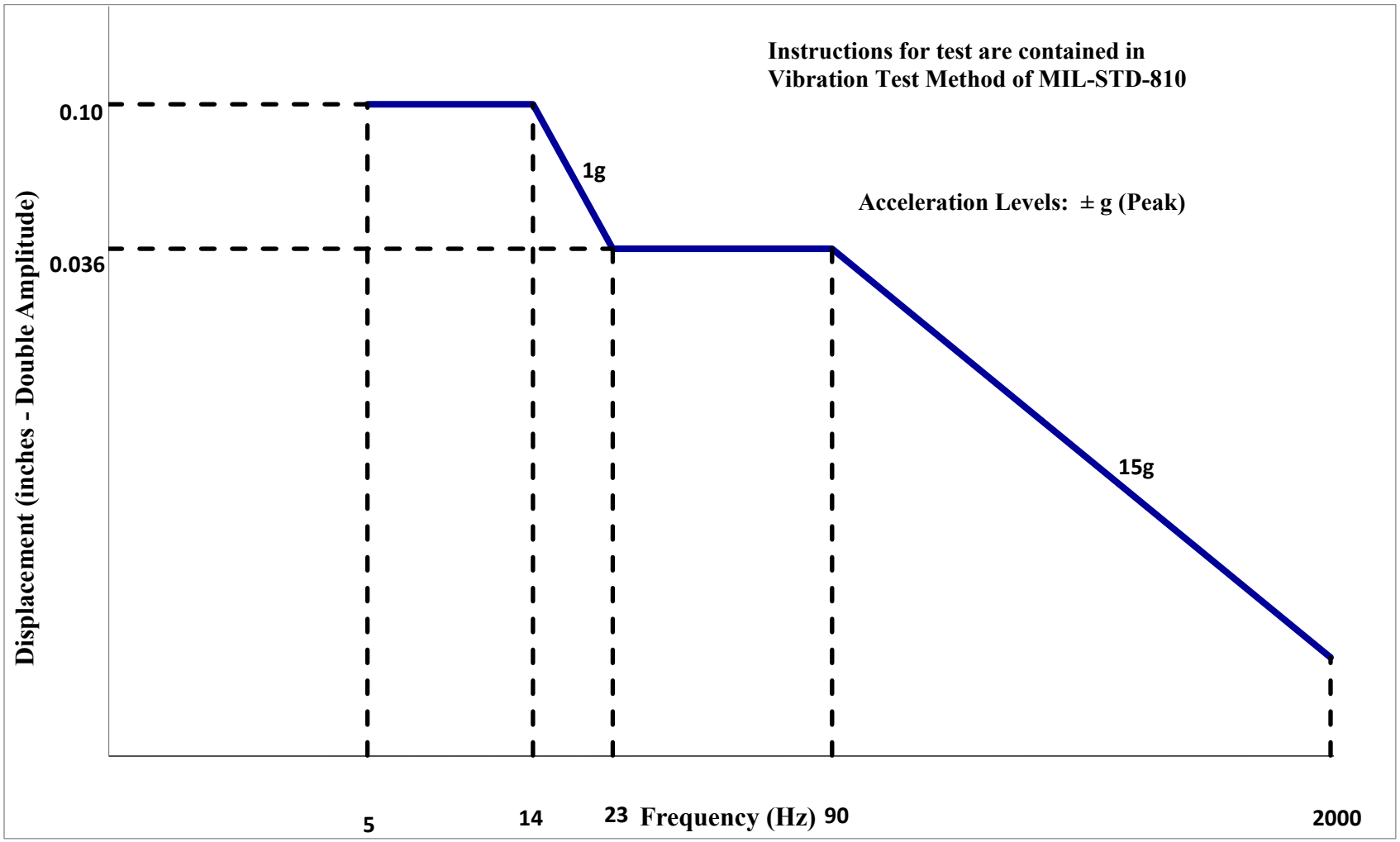


FIGURE 5. Vibration test curve.

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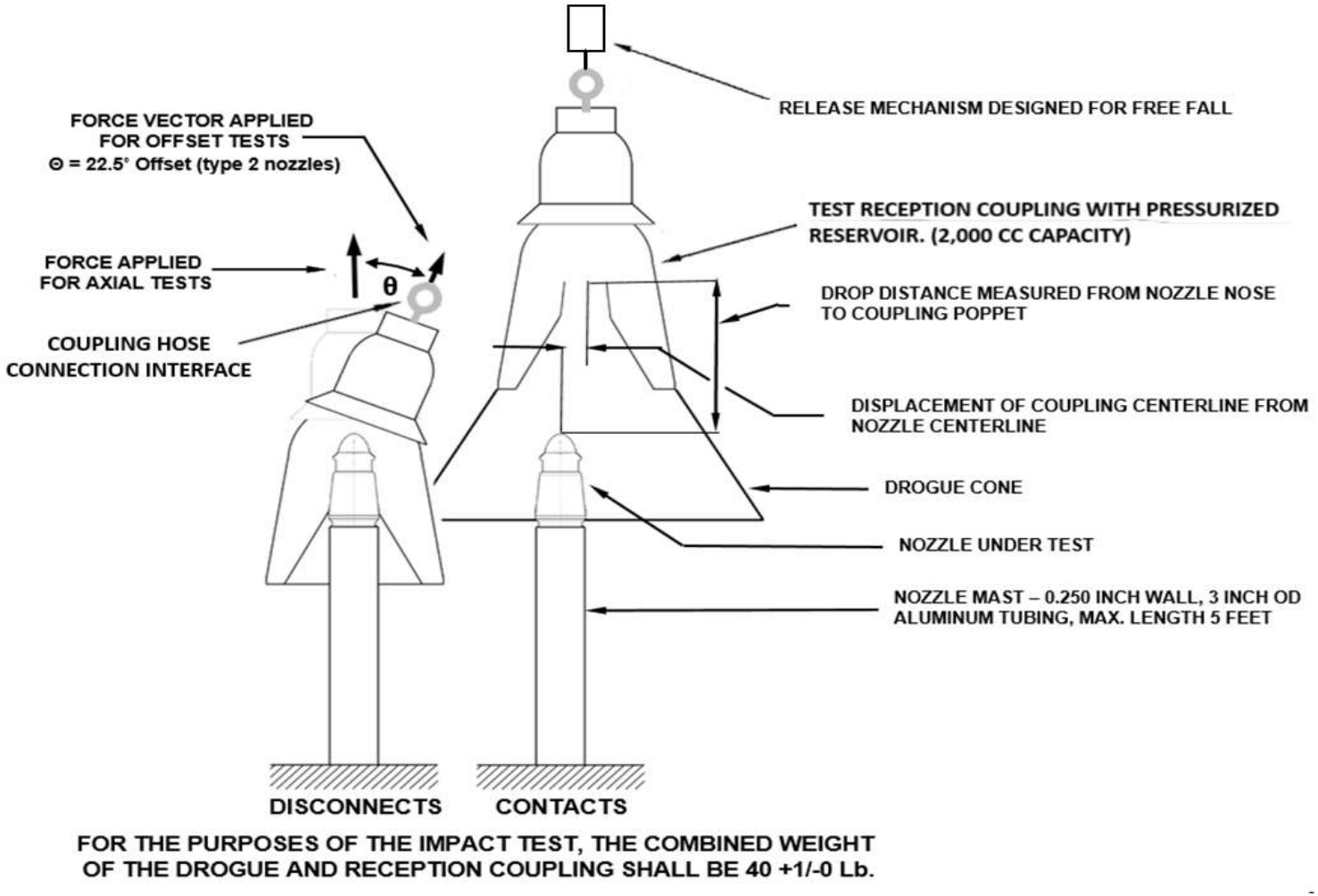


FIGURE 6. Nozzle test setup.

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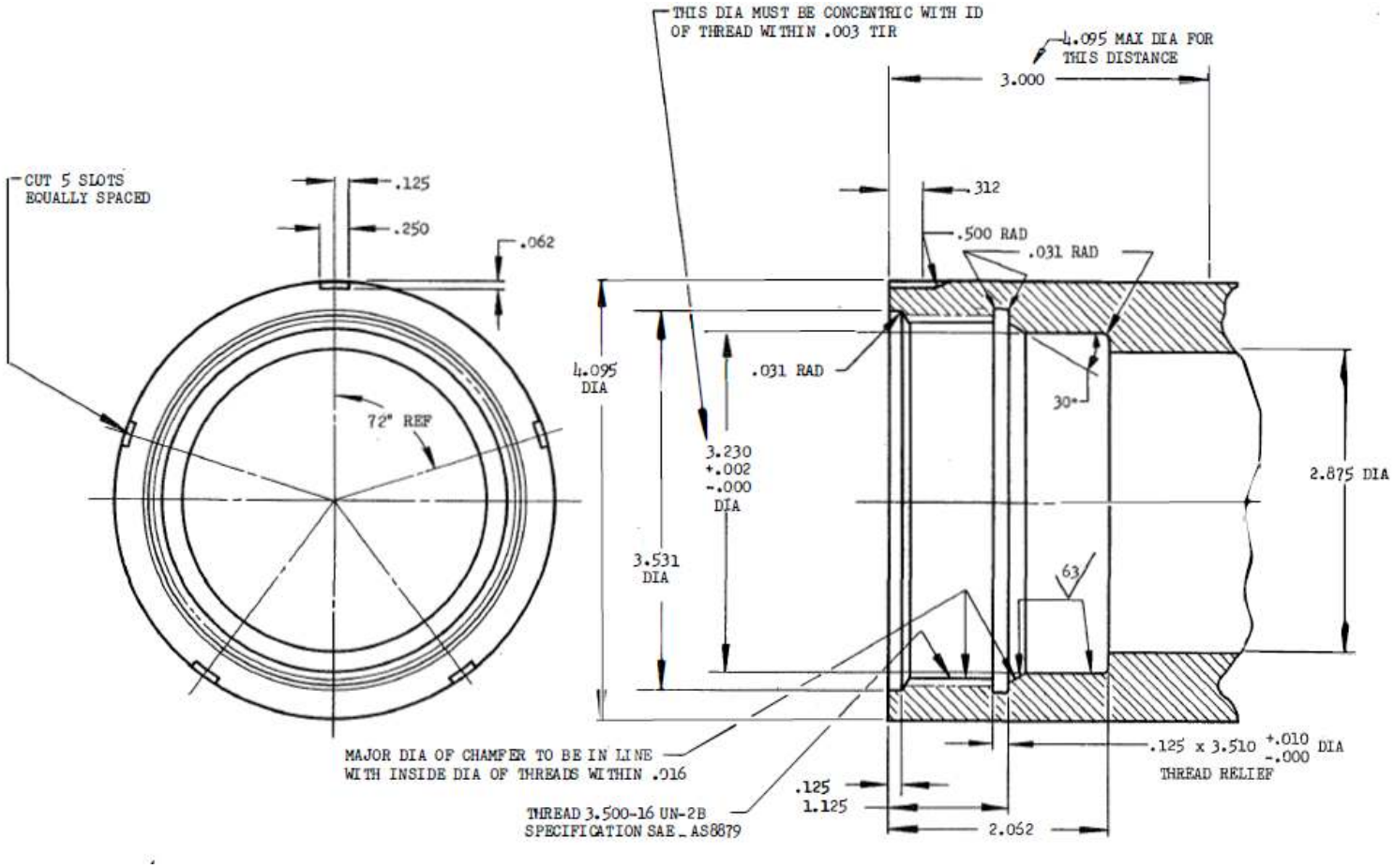
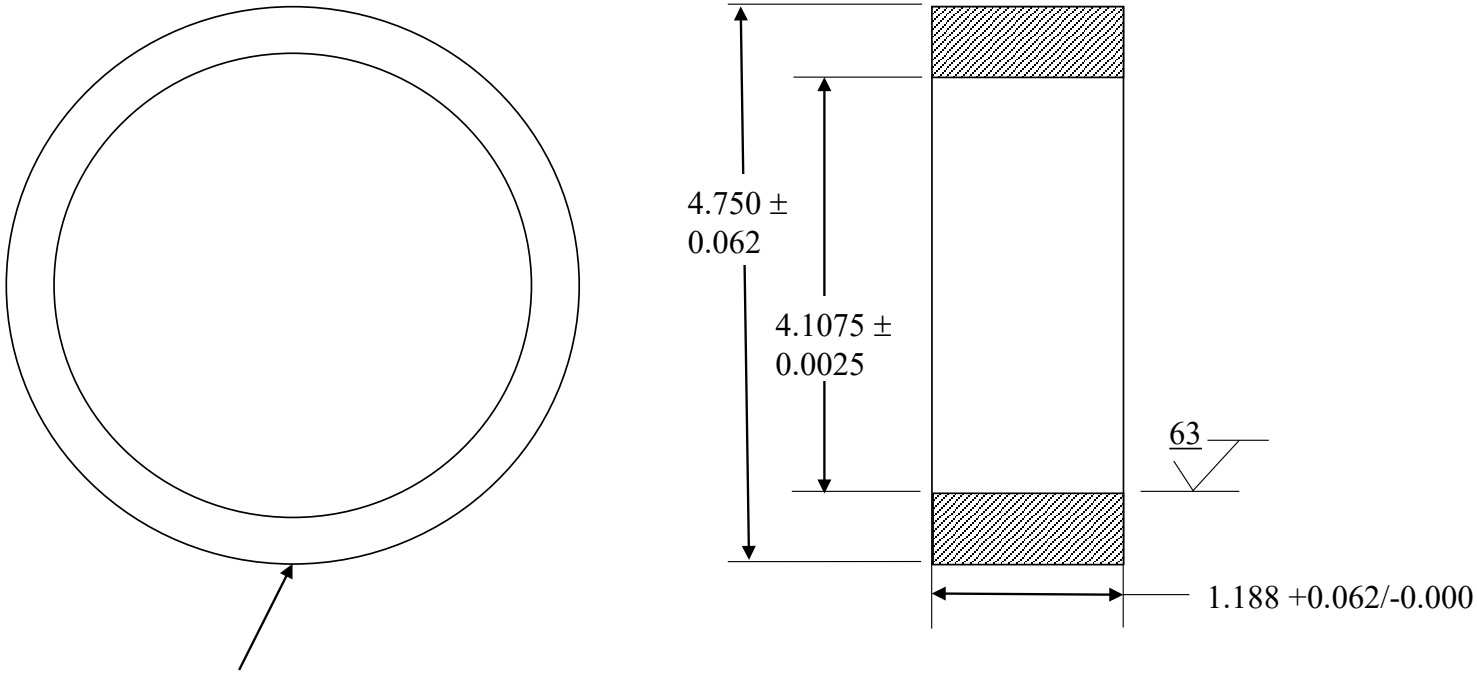


FIGURE 7. Probe mast interface dimensions.

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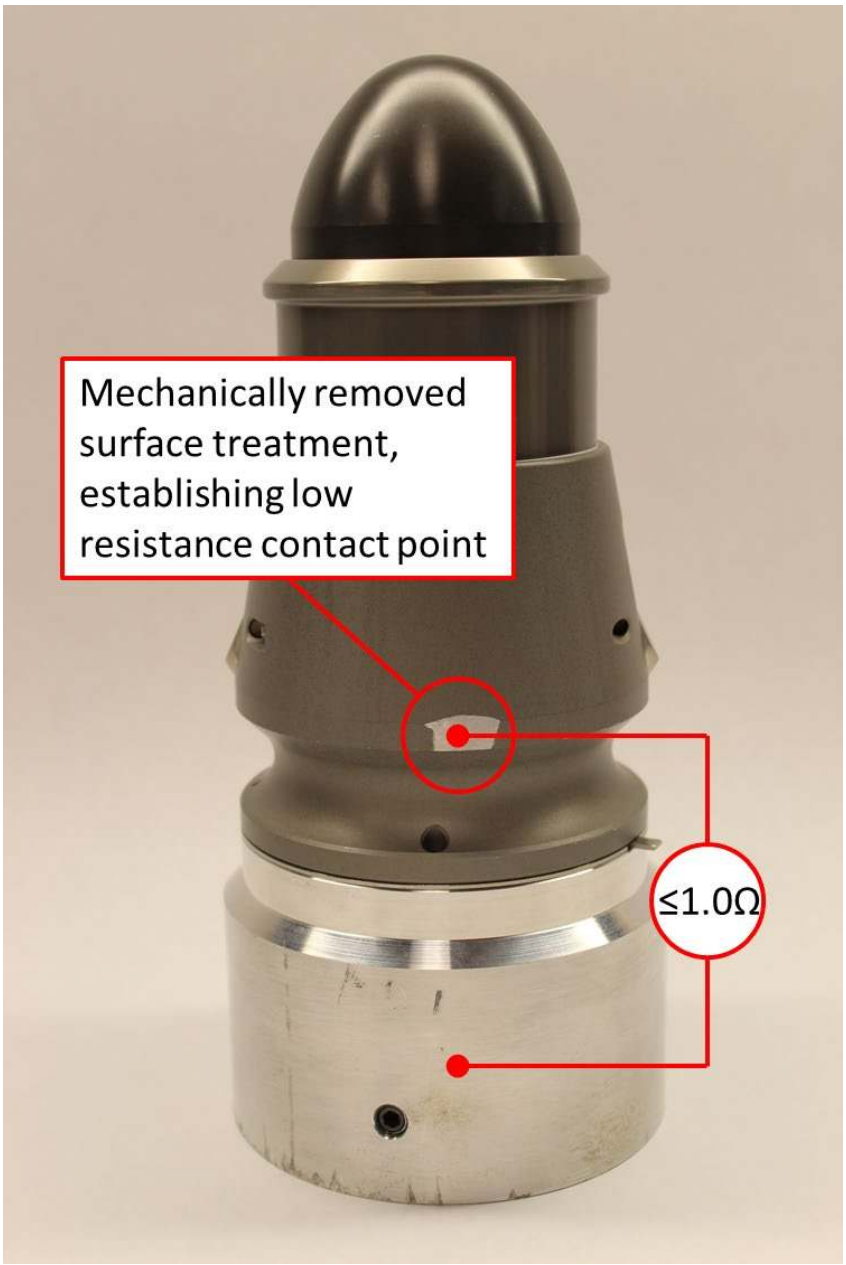
KNURL O.D. OR HANDLES OPTIONAL

NOTE: SURFACE TEXTURE IN ACCORDANCE WITH ANSI/ASME B46.1. BREAK SHARP EDGES 0.010 MAX.

MATERIAL: CORROSION RESISTANT STEEL.

FIGURE 8. Test gauge.

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NOTES:

1. Representative probe mast to be manufactured in accordance with MS24359, constructed from an aluminum alloy.

FIGURE 9. Electrical bonding resistance measurement.

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – AS
Air Force – 11
DLA – GS

Preparing activity:
Air Force – 11

(Project 1680-2024-001)

Reviewers:
Air Force – 71
DLA – GS3

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